# Sharath Naganna Honnavalli

2nd Year MS (Research), CSE, Indian Institute of Technology Bombay
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## Research Interests

Mechanistic interpretability, focused on understanding and controlling how models represent concepts internally. Particularly interested in whether the same concept shares an internal structure across notations, prompting regimes, and modalities.

## Education

**Indian Institute of Technology Bombay** MH, India
*Master of Science (by Research) in Computer Science and Engineering* 2024 – 2026 (Dec, expected)

- GPA: 9.48 / 10
- Advisors: Dr. Pushpak Bhattacharyya; Dr. Swaprava Nath; Dr. Aditya Joshi (UNSW, Australia)

**JSS Science and Technology University, Mysuru** Karnataka, India
*Bachelor of Engineering in Computer Science and Engineering* 2018 – 2022

- GPA: 9.51 / 10

## Publications

1. **Sharath Naganna**, Tanvir Ahmed Sijan, Uddipta Kalita. *"Are Arithmetic Heuristic Neurons Form-Invariant? A Mechanistic Analysis of Symbols, Text, and Code in LLMs."* Ninth Workshop on Analyzing and Interpreting Neural Networks for NLP (BlackboxNLP 2026). arXiv:2607.16693 — `[code]`
2. **Sharath Naganna**, Ritesh Sur Chowdhury, Dipankar Srirag, Aditya Joshi. *"Scaled Autoregressive Projection Steering (SAPS): Token-level Steering to Induce Sarcasm and Humblebrag in Text."* Findings of EMNLP 2026, Budapest.
3. **Sharath Naganna**, Saprativa Bhattacharjee, Biplab Banerjee, Pushpak Bhattacharyya. *"My life is miserable, have to sign 500 autographs everyday": Exposing Humblebragging, the Brags in Disguise.* Findings of ACL 2025, Vienna, Austria.
4. Akshata A. Bhat, **Sharath Naganna**, Saiful Haq, Prashant Khatri, Krishna Chaitanya Reddy Tamataam, Neha Arun, Niyati Chhaya, Pushpak Bhattacharyya. *"SAVIOR: Sample-efficient Adaptation of Vision-Language Models for OCR Representation."* IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops, 2026, pp. 719–728.

## Research Experience

**Scaled Autoregressive Projection Steering (SAPS) for Figurative Language**
*Indian Institute of Technology Bombay* (Dr. Aditya Joshi, UNSW) 2025–present

- Proposed SAPS, a residual-stream intervention that recomputes the steering coefficient at every decoding step so the intervention tracks incongruity as it unfolds, rather than fixing the coefficient at the first token as prior steering methods do.
- Analysed steering dynamics through layer-wise activation studies and token-level distribution divergence, improving grammaticality over fixed-coefficient steering baselines while preserving source content.

**Mechanistic Connection between Zero-Shot and In-Context Learning**
*Indian Institute of Technology Kanpur* (Dr. Ashutosh Modi) 2025–present

- Compared the sparse circuits underlying zero-shot and in-context learning using edge attribution patching with integrated gradients (EAP-IG), across six tasks and three model families (Llama-3.1-8B, Mistral-7B, Qwen2.5-7B).
- Measured circuit compressibility, cross-regime transfer, and structural overlap, finding asymmetric transfer: ICL circuits (median 5.95× larger) preserve zero-shot behaviour, while zero-shot circuits fail to recover ICL behaviour.
- Established that a median 86% of zero-shot circuit edges lie within the corresponding ICL circuit (3.59× above chance), evidence that ICL extends rather than replaces the zero-shot computation.

**Form-Invariance of Arithmetic Circuits in LLMs**
*Independent collaboration* 2026

- Investigated whether arithmetic heuristic neurons are form-invariant across symbolic arithmetic, natural-language word problems, and Python code in three Llama-3 models.
- Localized these neurons with a two-stage pipeline combining attribution patching and activation patching, applied separately to each input format.
- Found a compact shared neuron set across all three formats, with targeted interventions confirming it is both necessary and sufficient for late-layer arithmetic computation.

**Multilingual Visual Document Reranking**
*Flipkart–IITB collaboration* (Dr. Soumen Chakrabarti, IITB; Abhinav Anand, Flipkart) 2025–present

- Built a multilingual dataset for visual document reranking, 250K queries across 6 languages with hard negatives mined from a multi-vector retriever (public release forthcoming).
- Designed the data pipeline, including a difficulty curriculum that downsamples easy queries, cutting rank-1 queries from 69% to 21%, and NLLB-200 translation with round-trip quality checks.
- Trained a 1B reranker that outperforms the 2B DocReRank across all 8 UniDoc-Bench domains at half the size, benchmarked against 5 rerankers on ViDoRe v3 and UniDoc-Bench.

**Sample-efficient Adaptation of Vision-Language Models for OCR**
*Hyperbots, Bengaluru* (Dr. Niyati Chhaya) May 2025–July 2025, June 2026–Aug 2026

- Proposed SAVIOR, a sample-efficient data curation strategy for adapting vision-language models to financial OCR, with a train set of 2,234 documents and a 509-document expert-annotated benchmark targeting failure modes such as vertical text, fine print, and complex layouts.
- Introduced PaIRS, a structure-aware metric measuring pairwise spatial relationships between tokens to evaluate layout fidelity beyond word accuracy; fine-tuned Qwen2.5-VL with LoRA, reaching 0.93 word recall and 0.89 F1 on downstream document QA.

## Industry Experience

**Software Engineer II** — *Cisco* Jan 2022–July 2024

- Progressed from Technical Intern to Software Engineer II in the Multi-Cloud Management team.
- Built backend services in Go and ETL pipelines between MongoDB and MySQL, with AWS, Azure and GCP account integration.

## Technical Skills

**Languages:** C, Python, Golang, SQL
**Frameworks / Tools:** PyTorch, Transformers, TransformerLens, vLLM, Flask

## Awards and Achievements

- Selected for the Heidelberg Laureate Forum (HLF), 2026.
- Department Travel Grant to attend ACL 2025.
- All India Rank 325 of 123,967 candidates, Graduate Aptitude Test in Engineering (GATE) in Computer Science, 2025 (national postgraduate entrance examination).
- Best Bachelor's Thesis (Research Category), JSS Science and Technology University.

## Service

- Volunteer, AACL-IJCNLP 2025, Indian Institute of Technology Bombay.